

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 1 of 9

Research Program/Project/Study Title: Development and Evaluation of Flavored Salted Egg	
Proponent Name:	Melinda O. Conlu, EdD
Academic Rank:	Associate Professor I
Co-Authors (with academic rank):	Ace Barredo, EdD, Asso. Prof II
Campus/Satellite College/Address:	CapSU Roxas City- Main Campus
Telephone/Cell Phone/Fax Number:	09686476531
E-mail Address:	Melayoroconlu.22@gmail.com

Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
		Municipality/City:	Roxas City						
		Province:	Capiz						
		Region:	VI						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☐ Basic	☐ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
☐ Basic	☐ Pilot Testing								
☒ Applied	☐ Tech. Promotion/Commercialization								
(7) Sector/Commodity Agriculture/Fruit and Vegetables		(8) Discipline/Field of Specialization Food Development							

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 2 of 9

(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

Salted duck eggs and preserved duck eggs (also known as ‘century eggs’ or ‘thousand yr old eggs’) were originally developed in China centuries ago to extend the shelf life of duck eggs in the absence of refrigeration (Hou, 1981).

The purpose of the study was to develop a flavored salted egg using rosemary, basil, and pandan leaves as natural flavoring agent. Locally available ingredients were utilized in making the flavored salted eggs. The product formulation underwent three trials with three treatments for the sensory evaluation. The best formulation of flavored salted eggs was evaluated for the consumers' acceptability. Likewise, the shelf life of flavored salted eggs was observed in order to ensure the duration of storage and expiration both at room and chilling temperature.

The study was an experimental-developmental method. The statistical tools used in the analysis and interpretation of data were the Mean and the Analysis of Variance (ANOVA) set at 0.01 alpha level. The flavored salted egg had shown to be liked extremely by 10 semi-trained panelists. On the other hand, consumer preferences on the acceptability of the flavored salted egg were liked extremely in terms of color, aroma, flavor, and texture considering the three flavoring leaves such as pandan, rosemary, and basil. Moreover, the storage of flavored salted egg at room temperature was less than 2 months, and for the chilling temperature, it stayed for less than 3 months without compromising the food quality.

Keywords: favored salted egg, pandan, rosemary and basils

(10) Introduction

People of today's generation are very fond of tasting new genres of food that satisfy them. Filipinos are craving for something new and unique and most especially if it is very beneficial. In this modern age, the old get old and the new is being patronized by everybody.

The Philippines is very rich in natural raw materials, however, it depends on how the man uses these in the right way. We are abundant in ingredients that can be found in our surroundings. There are unknown useful materials that can be used as an alternative or to elevate some other products that will make its taste more fascinating. Maximizing the use of our raw materials can lift up our economy because the discovery of a new product will be altered into money in the easiest way.

This study was conducted in order to maximize the use of our raw materials, so the researcher was keen to find a solution to raise our unbalanced economy. The implication of innovative ideas, skills, and scientific methods will produce a fruitful method.

This research is entitled as “Acceptability of Flavored Salted Eggs with Rosemary, Basil, and Pandan Leaves”. This research will try to intoxicate the salted egg into a more palatable and interesting one with the use of rosemary, basil, and pandan leaves that will act as a flavoring. The three varieties of plants can be found in our surroundings, especially in cold places and are known for its benefits because of its exemplary uses and nutritive value that is very useful in the health of human.

The importance of this study is to inform the people that the Basil, Rosemary, and Pandan leaves can be used as flavoring. This will also inform and encourage them to plant and care for Basil, Rosemary, and Pandan.

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 3 of 9

This study generally aimed to:

1. determine the sensory qualities of flavored salted egg in terms of color, aroma, flavor and texture among three treatments;
2. find out the acceptability of flavored salted egg in terms of appearance, aroma, taste and texture among three treatments;
3. find out the storage quality of flavored salted eggs in terms of temperature.

Duck eggs are notable because they're almost 50% larger than a large-sized hen's egg. They have a large, golden, creamy yolk, and many people love them for their rich, extra-egg flavor. Their shells are also a treat for the eyes. Compared with white or brown chicken eggshells, duck eggs come in a range of colors, including pale blue, blue-green, charcoal gray, and occasionally white.

Salted duck eggs and preserved duck eggs (also known as 'century eggs' or 'thousand yr old eggs') were originally developed in China centuries ago to extend the shelf life of duck eggs in the absence of refrigeration (Hou, 1981). These egg products are still widely consumed today, especially in Asia, and are key ingredients in a variety of East and Southeast Asian dishes. Although chicken eggs are typical in the Western diet with a worldwide production of 1.249 trillion shell eggs in 2012, worldwide production of shell eggs from other poultry species saw a 32% increase from 2002, exceeding 87 billion in 2012 with 94% of production concentrated in Asia1 (FAO Statistics Division, 2014a,b).

Salted eggs are native to Chinese and Filipino cuisines. The process traditionally calls for duck eggs, but you can use chicken eggs when duck eggs are unavailable. You can consume the finished eggs alone after boiling them, but you can also use salted eggs as ingredients for moon cakes and other foods.

Pandan is a herbaceous tropical plant that grows in Southeast Asia. This plant family comprises about 250 species of evergreen trees, shrubs, and scramblers (Bown, 2002). Many of these plants are grown for their architectural appearance, either as landscape plants or as ornamentals under cover, whereas most species can be found in the forest and some are coastal plants. The plants in this family are often known as 'screw pine' because they resemble Ananas plants (pineapple) with a spiral arrangement of long, narrow, and strap-shaped leaves. Leaves of some species have a toothed edge along their margin. Among these various species, there are two types of fragrant screw pine, Pandanus leaves (*Pandanus amaryllifolius*) and Pandanus flowers (*Pandanus odoratissimus*), which are divided according to the part of plant that bears scent. *P. amaryllifolius* or pandan wangi is the only Pandanus species with fragrant leaves. It is a short shrub of 1.2–1.5 m (4–5 ft) in height and 60–90 cm (24–36 inch) in width with a stout stem and usually branched low down. Their aromatic, linear, pointed leaves, with no toothed edge, are about 80 cm (32 inch) long and 5 cm (2 inch) wide (Bown, 2002). The plant never flowers, thus the fruits are unknown. Natural distribution is found over Southern India, the Southeast Asia peninsular, Indonesia and Western New Guinea. Nowadays, it is well-known as a characteristic herb of Southeast Asia cuisines, in which its leaves are mainly used as food flavorings.

Rosemary is a member of the mint family *Lamiaceae*, along with many other herbs, such as oregano, thyme, basil, and lavender. The herb not only tastes good in culinary dishes, such as rosemary chicken and lamb, but it is also a good source of iron, calcium, and vitamin B-6. It is typically prepared as a whole dried herb or a dried powdered extract, while teas and liquid extracts are made from fresh or dried leaves. The herb has been hailed since ancient times for its

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 4 of 9

medicinal properties. Rosemary was traditionally used to help alleviate muscle pain, improve memory, boost the immune and circulatory system, and promote hair growth.

Basil is used in traditional Tamil and Ayurvedic medicine, which is a form of traditional medicine popular on the Indian subcontinent. There are a number of types of basil, which differ in taste and smell. Sweet basil (the most commercially available basil used in Italian food) has a strong clove scent because of its high concentration of the chemical agent eugenol. Alternatively, lime and lemon basil have a strong citrus scent due to their high concentration of limonene. Basil comes in many varieties. While it may not contribute significant vitamins and nutrients to your diet, it can add unique flavor to your sauces, salads, and soups. Basil is a flavorful, leafy green herb that originated in Asia and Africa. It's a member of the mint family, and many different varieties exist. Popular as a food seasoning, this aromatic herb is also used in teas and supplements which may provide a range of health benefits. A member of the mint family (Lamiaceae), basil is one of the most popular culinary herbs. It's a warm-weather annual herb you can plant outdoors once temperatures are consistently above 50°F. Basil is not usually seeded directly into the soil; typically, gardeners transplant small starter plants purchased at a nursery. Or, they may start the seeds themselves indoors under grow lights.

Related Studies:

Salted egg, a traditional characteristic processed egg product in China, is popular among consumers at home and abroad. Salted egg quality characteristics formation primarily includes the hydration of egg white, the solidification of egg yolk, the unique color and flavor of salted egg yolk, and the formation of white, fine, and tender egg whites and loose, sandy, and oily egg yolks after pickling and heating. The unique quality characteristics of salted eggs are mostly caused by the infiltration dehydration of salt, the intermolecular interaction of proteins, and the oxidation of lipids. In recent years, to solve the problems of salted eggs having high salinity, long production cycle, and short storage period, the pickling technology for salted egg has been improved and researched, which has played a significant role in promoting the scientific production of salted eggs. This paper summarizes the mechanisms of salted egg quality characteristics formation and factors influencing quality, with a perspective of providing a theoretical basis for the production of high-quality salted eggs.

(11)

Procedure in the Preparation of ingredients of flavoured salted egg. Prepare the tools and equipment needed. Wash and dry the eggs, sterilize the container or jar, place the egg in the jar, and the three leaves basil, rosemary and pandan. Wash and cut the leaves. Set aside for later use

Procedure in the Preparation brine solution . Boil water gradually add the salt and the leaves(basil, rosemary, and pandan).

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 5 of 9

1. Procedure in the Preparation salting the egg.

1. Boil 500 ml of water and 250 grams of salt with 100 grams, pandan, basil, and rosemary leaves.
2. Carefully place the duck eggs in a wide smooth glass jar.
3. Pour the brine solution together with pandan, basil and rosemary leaves into a jar. Weight down the eggs with plastic with saucer to keep.
4. Lower the mouth of the jar, keep jar in a cool dry place.
5. Try one egg after 18 days soak the eggs again for three days more if tried eggs is not salty enough
6. then After twenty one days harvest the eggs and bring to boil the salted egg for 15 minutes.
7. Flavored salted egg

The Nine (9) Point Hedonic rating scale was used to determine the sensory qualities in terms of color, aroma, flavor, and texture, and acceptability of the product.

It dealt with three treatments and trials. Trial 1, 50 gms, 75 gms, and 100 gms of basil, rosemary and pandan leaves. Several trials were done to come up with the most acceptable product in terms of sensory qualities such as color, aroma, flavor, and texture

1. Production of technology manual/cooking guide/GMP manual

The quality standards, cooking guide, and other IEC materials will be developed and produced. It will be disseminated and distributed to various interested stakeholders.

The experimental design will be CRD with 3 treatments. The gathered data will be computed using the SPSS and analyzed using Analysis of Variance (ANOVA) at a 0.01 percent level of significance.

(12) Results and Discussions

Sensory Evaluation of Flavored Salted Egg

Table 1 shows the sensory evaluation of flavored salted egg in terms of color, aroma, flavor, and texture using three treatments. The three leaves as flavoring agent of salted egg were evaluated by the semi-trained panelists. Results showed that treatment C (100 grams of flavoring leaves) of salted eggs got the highest results in terms of color, aroma, flavor, and texture.

This implies that 100 grams of pandan, basil, and rosemary leaves as a flavoring agent for salted eggs was the right proportion.

Table 1. Sensory qualities of flavored salted egg in terms of color, aroma, flavor and

texture.

	BASIL LEAVES			ROSEMARY LEAVES			PANDAN LEAVES		
Sensory Qualities	A (50)	B (75)	C (100)	A (50)	B (75)	C (100)	A (50)	B (75)	C (100)
Color	8.3	8.9	8.9	8.6	8.5	8.7	8.9	8.8	9.0
Aroma	7.4	7.3	7.9	7.5	7.7	8.2	8.9	9.0	8.9
Flavor	8.9	8.5	8.9	8.8	8.5	8.7	8.7	9.1	9.2
Texture	8.2	8.5	8	8.9	8.8	9.0	8.6	8.9	9.1

General Acceptability of Flavored as Evaluated by Salted Egg as evaluated by Consumers

Table 2 reflect the consumer evaluation on the acceptability of flavored salted egg shows that in terms of color treatment C got the mean score of 8.45 interpreted as liked extremely, in terms of aroma treatment C got the mean of 8.48, in terms of flavor treatment C got the mean score of 8.45 and in terms of texture treatment c got the mean score of 8.45.

Table 2. General Acceptability of Flavored as Evaluated by Salted Egg as evaluated by Consumers.

TREATMENTS	A		B		C	
<i>Quality Attributes</i>	<i>Mean</i>	<i>QD</i>	<i>Mean</i>	<i>QD</i>	<i>Mean</i>	<i>QD</i>
color	8.20	LE	8.25	LE	8.45	LE
Aroma	8.38	LE	8.28	LE	8.48	LE
flavor	8.35	LE	8.35	LE	8.65	LE
Texture	8.15	LE	8.18	LE	8.45	LE
Acceptability	8.27	LE	8.26	LE	8.51	LE

Legend:

Scale of Means

8.12 – 9.00

7.23 – 8.11

6.34 – 7.22

Qualitative Description

Liked Extremely

Liked Very Much

Liked Moderately

Storage Quality of Flavored Salted Egg

The shelf-life of flavored leaves in three (3) variants, stored in a cool, dark, and dry area at room temperature not more than 32°C. The data shows that less than 2 months of observation, there was no mold formation, unpleasant odor, slimy, and production of the spot which means

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 7 of 9

the product was still safe for human consumption. Thus, the nutritive value, as well as the flavor and appearance would be best protected by low temperature and low humidity during storage (Villarruz, 2015).

Table 2 .Shelf-life of the flavored salted egg at RoomTemperature.

Variant	One-month Mold Formation, unpleasant odor	One and half months Mold Formation unpleasant odor	Two-months Mold Formation, unpleasant odor, production of spot
Treatment A	0	0	+
Treatment B	0	0	+
Treatment C	0	0	+

0 – absence of molds

+ – presence of molds

However, when stored at a chilling temperature for less than three (3) months, the flavored salted egg showed no change in the preparation during the first to two months of storage. But, during its three (3) months of storage, changes like mold, and unpleasant odor were observed (Villaruzz, 2015)

Table 3. Shelf-life of the flavored salted egg at Chilling Temperature.

Variant	One-month Mold Formation, unpleasant odor	One and half- months Mold Formation, unpleasant odor	Three-months Mold Formation unpleasant odor, production of spot
Treatment A	0	0	+
Treatment B	0	0	+
Treatment C	0	0	+

(13) Conclusions and Recommendations

Conclusions

Based on the findings of the study, the following conclusions are drawn:

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 8 of 9

Salted eggs can be flavored using three kinds of leaves, such as pandan, basil, and rosemary.

On the other hand, consumers' preferences are based on the acceptability level the pandan is the first, followed by basil, and the last rosemary considering the sensory qualities.

The storage of flavored salted egg at room temperature was less than 2 months, and for the chilling temperature, it stayed for less than 3 months without compromising the food quality.

Recommendations

Based on the conclusions of the study, the following recommendations are drawn:

Pandan leaves, basil leaves and rosemary leaves are recommended in making flavored salted eggs.

It is further recommended to housewives, consumers, home economics, and food technologists to make and use leaves (pandan, basil, rosemary) as flavored to a salted egg.

Further study may be conducted considering other variables not included in this study.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

Flavored salted can be produced and can be commercialized

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

The product of this study could be one of the sources of extension activities conducted by the University, particularly on the main campus.

(16) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

References

Elsevier B.V. 2023

Foods. 2022 Oct; 11(19): 2949.

Published online 2022 Sep 21. doi: 10.3390/foods11192949

Medically reviewed by Miho Hatanaka, RDN, L.D. — By Anne Danahy, MS, RDN on June 18, 2020

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 9 of 9

Yankee Publishing Inc., 2023

Villaruz , 2015

H.C. Hou (Hou Xiangchuan) 1981. Egg preservation in China. Food Nutr. Bull., 3(2) (1981)
 AccessedMar.2012 [http://archive.unu.edu/unupress/food/8F032e/8F032E03.htm#Eggpreservati](http://archive.unu.edu/unupress/food/8F032e/8F032E03.htm#EggpreservatinChina)
 oninChina